

Rithika Ravi	P575	B01	"Week 1 — C# Language & Object Foundations".
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Day 3 — Memory Model, Enums, Nullable, Conversion

Task 1.9 Copy-and-modify test with two ints, then an int[], then a Coordinate struct vs a Coordinate class. Explain each output in a comment. ✓ Done when: the comment correctly explains why the array/class shared state but the int/struct did not.

Procedure:

From this exercise I have come to know that int and struct are value types meaning if we modify the value after copying, it doesn't change the original value. But in reference types like classes and array, we can't modify the copied value because the copied value holds a pointer to the address of data instead of actually storing the data. Hence, the original value will also be modified.

```

11 }
12
13 class Program
14 {
15     static void Main()
16     {
17         //Copying and modifying two ints
18         int a = 5;
19         Console.WriteLine($"Value of a: {a}");
20         int b = a; //Copying the value of a to b
21         Console.WriteLine($"Copying value of a to b: {b}");
22         a = 6; //Changing value of a after copying to b
23         Console.WriteLine($"Changing value after copying ");
24         Console.WriteLine($"Value of a: {a}");
25         Console.WriteLine($"Value of b: {b}"); //b remains unchanged
26
27         //Changing two array values
28         Console.WriteLine("Changing two array values");
29         int[] arr1 = { 1, 2, 3 };
30         int[] arr2 = arr1; //Copying the reference of arr1 to arr2
31
32         //Changing value of arr2
33         arr2[0] = 11;
34         Console.WriteLine($"New value of arr2[0]: {arr2[0]}");
35         Console.WriteLine($"Reference change of arr1[0]: {arr1[0]}"); //arr1 also changes because arr2 is a reference to arr1
36
37         //Value type: Struct
38         Point p1 = new Point { X = 5 };
39         Point p2 = p1; // Copying the value of p1 to p2
40
41         p2.X = 100;
42         Console.WriteLine($"Value of p1.X: {p1.X}"); //unchanged because p2 is a copy of p1
43         Console.WriteLine($"Value of p1.X: {p2.X}"); // changed because p2 is a copy of p1
44
45         //Reference type: Class
46
47         Person person1 = new Person { Age = 30 };
48         Person person2 = person1; // same reference
49
50         person2.Age = 50;
51         Console.WriteLine($"Value of person1.Age: {person1.Age}"); // 30 (changed)
52         Console.WriteLine($"Value of person2.Age: {person2.Age}"); // 50
53
54     }
55 }
56
57

```

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```

Microsoft Visual Studio Debug Console
Value of a: 5
Copying value of a to b: 5
Changing value after copying
Value of a: 6
Value of b: 5
Changing two array values
New value of arr2[0]:11
Reference change of arr1[0]:11
Value of p1.X:5
Value of p1.X:100
Value of person1.Age: 50
Value of person2.Age: 50

C:\Users\ravi.rithika\source\repos\DayThreeTask\DayThreeTask\bin\Debug\net10.0\DayThreeTask.exe (process 13676) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```

Task 1.10 DaysOfWeek enum: take a number 0–6, print the day name. [Flags] FilePermission {Read=1, Write=2, Execute=4}: combine with |, test with &. ✓ Done when: combined permissions print correctly and a single permission check works.

Procedure:

Created a Days of week enum and took number as 6 then printed the day name.

```

Program.cs
DayThreeTask
using System;

struct Point
{
    public int X;
}

class Person
{
    public int Age;
}

enum DaysOfWeek
{
    Sunday,
    Monday,
    Tuesday,
    Wednesday,
    Thursday,
    Friday,
    Saturday
}

class Program
{
    static void Main()
    {
        int number = 6;
        DaysOfWeek day = (DaysOfWeek)number;

        Console.WriteLine(day); // Output: Friday
        Console.WriteLine((int)day); // Output: 5
    }
}

```

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```

Microsoft Visual Studio Debu... X + -
Saturday
6
C:\Users\ravi.rithika\source\repos\DayThreeTask\DayThreeTask\bin\Debug\net10.0\DayThreeTask.exe (process 12204) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```

Created an enum named file permission. Combined with | and & tested in different scenarios to display output.

```

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search DayThreeTask
Debug Any CPU DayThreeTask
Program.cs
DayThreeTask
22 }
23
24
25 [Flags]
26 5 references
27 enum FilePermission
28 {
29     Read = 1,
30     Write = 2,
31     Execute = 4
32 }
33
34 0 references
35 class Program
36 {
37     0 references
38     static void Main()
39     {
40
41         FilePermission perms = FilePermission.Read | FilePermission.Write;
42
43         Console.WriteLine(perms); // Read, Write
44
45         bool hasRead = (perms & FilePermission.Read) != 0;
46         bool hasExecute = (perms & FilePermission.Execute) != 0;
47
48         Console.WriteLine(hasRead); // true
49         Console.WriteLine(hasExecute); // false
50
51
52
53
54
55
56
57
58
59
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61
62
63
64
65
66

```

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```

Microsoft Visual Studio Debug Console
Read, Write
True
False

C:\Users\ravi.rithika\source\repos\DayThreeTask\DayThreeTask\bin\Debug\net10.0\DayThreeTask.exe (process 26816) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```

Task 1.11 Nullable int with HasValue; method ApplyDiscount(double? discount) defaulting to 5% via ??.

✓ Done when: passing null applies the default; passing a value uses it.

Procedure:

Here, nullable int with HasValue method returns false. Using null coalescing(??) we can assign a default value if the value is null. Created a applydiscount method where default 5% is applied if null value is passed and if proper discount rate is passed it uses that value.

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The image shows a screenshot of the Visual Studio Code editor with the following components:

- Top Menu Bar:** File, Edit, View, Git, Project, Build, Debug, Test, Tools, Extensions, Window, Help, Search, DayThreeTask.
- Toolbar:** Includes icons for undo, redo, save, open, close, run and debug, and other development utilities.
- Editor Tabs:** Program.cs (active), DayThreeTask, and ApplyDiscount(double? discount).
- Code Content:**

```
<pre><code>24
25 [Flags]
26 0 references
27 enum FilePermission
28 {
29     Read = 1,
30     Write = 2,
31     Execute = 4
32 }
33
34 0 references
35 class Program
36 {
37     2 references
38     static void ApplyDiscount(double? discount)
39     {
40         double actual = discount ?? 0.05; // default 5%
41         Console.WriteLine($"Applying {actual * 100}% discount");
42     }
43
44     0 references
45     static void Main()
46     {
47         int? age = null;
48
49         Console.WriteLine(age.HasValue); // false
50         Console.WriteLine(age ?? 18); // 18 (default)
51
52         age = 25;
53         Console.WriteLine(age.HasValue); // true
54         Console.WriteLine(age.Value); // 25
55
56         ApplyDiscount(null);
57         ApplyDiscount(0.10);
58
59
60
61
62
63
64</code></pre>
```

```
False
18
True
25
Applying 5% discount
Applying 10% discount

C:\Users\ravi.rithika\source\repos\DayThreeTask\DayThreeTask\bin\Debug\net10.0\DayThreeTask.exe (process 20148) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .|
```

Task 1.12 Show `int`→`long`→`float`→`double` (implicit) and `double`→`int` (explicit, precision loss). Take a string and use `is`, `as`, `Convert.ToInt32`, `int.TryParse`, noting when each is right. ✓ Done when: all four conversion approaches run and the precision-loss case is noted.

Procedure:

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Learned different ways C# converts data: implicit (safe automatic), explicit casting (manual with possible data loss), type checking (is/as), and string conversions using Convert and TryParse.

- `is` → "Is this type?" → **true/false**
- `as` → "Try to convert" → **result or null**

```

44 // Program
45 static void Main()
46 {
47     Console.WriteLine("==== Implicit Conversion (safe widening) =====");
48
49     int i = 42;
50     long l = i; // int to long
51     float f = i; // int to float
52     double d = i; // int to double
53
54     Console.WriteLine($"int: {i}, long: {l}, float: {f}, double: {d}");
55
56     Console.WriteLine("==== Explicit Casting (possible data loss) =====");
57
58     double d2 = 3.99;
59     int i2 = (int)d2; // truncates decimal
60
61     Console.WriteLine($"double: {d2} to int: {i2} (decimal lost)");
62
63     Console.WriteLine("==== 'is' and 'as' (reference types) =====");
64
65     object obj = "Hello";
66
67     // using 'is'
68     if (obj is string s)
69     {
70         Console.WriteLine($"is success: {s}");
71     }
72
73     // using 'as'
74     string s2 = obj as string;
75     Console.WriteLine($"as result: {s2}");
76
77     Console.WriteLine("==== Convert (throws if invalid) =====");
78
79     int x = Convert.ToInt32("42"); // valid conversion
80     Console.WriteLine($"Convert.ToInt32(\"42\") = {x}");
81
82     Console.WriteLine("==== TryParse (safe, no exception) =====");
83
84     bool ok = int.TryParse("abc", out int y);
85     Console.WriteLine($"TryParse(\"abc\") success: {ok}");
86     Console.WriteLine($"Parsed value: {y}");
87 }
88
89
90
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107
108
109
110

```

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```
Microsoft Visual Studio Debug Console
=== Implicit Conversion (safe widening) ===
int: 42, long: 42, float: 42, double: 42

=== Explicit Casting (possible data loss) ===
double: 3.99 ↗ int: 3 (decimal lost)

=== 'is' and 'as' (reference types) ===
'is' success: hello
'as' result: hello

=== Convert (throws if invalid) ===
Convert.ToInt32("42") = 42

=== TryParse (safe, no exception) ===
TryParse("abc") success: False
Parsed value: 0

C:\Users\ravi.rithika\source\repos\DayThreeTask\DayThreeTask\bin\Debug\net10.0\DayThreeTask.exe (process 10980) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .|
```